

Inside atoms

ATOMS ARE TOO SMALL TO SEE, EVEN WITH SOME OF THE MOST POWERFUL MICROSCOPES.

Everything we can see in the Universe, from the stars to our own bodies, is made up of atoms.

SEE ALSO

◀ 98–99 States of matter

◀ 108–109 Elements and atoms

◀ 116–117 The periodic table

◀ 126–127 Radioactivity

Forces and mass 172–173 ▶

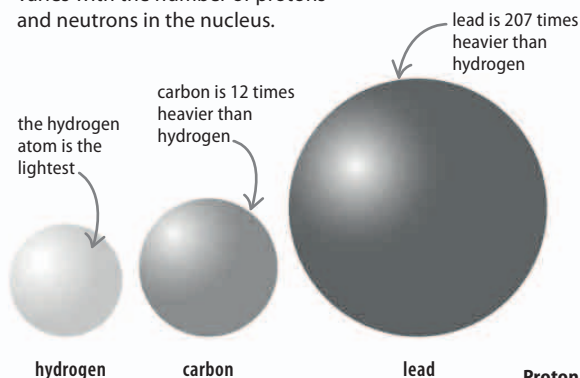
Electricity 202–203 ▶

What is an atom?

Atoms are not all the same. There are 92 different types that occur naturally—and a few more that are made by scientists in laboratories. Each atom belongs to a specific element, a substance that cannot be purified further into simpler ingredients. Familiar elements include hydrogen, carbon, and lead.

▽ Different atoms

The atoms of every element have a unique size and mass. The mass varies with the number of protons and neutrons in the nucleus.



Subatomic structure

Atoms are made up of even smaller particles called protons, neutrons, and electrons. The atoms of a certain element have a unique number and arrangement of particles, which is what gives the element its distinct properties—making it a gas or metal, for example. An atom always has the same number of protons as electrons. Each proton has a positive charge, which is matched by the negative charge of an electron, making the whole atom neutral.

Nucleus

The protons and neutrons form the nucleus, a tiny core where most of the matter is packed.

Proton

Protons have positive charges that attract the negatively charged electrons, holding them in place around the nucleus.

Neutron

These particles have no charge. They make up the rest of the mass of the atom, each weighing slightly more than a proton.

△ Carbon atom

All carbon atoms have six protons in the nucleus and an equal number of electrons moving around it. Most carbon atoms also have six neutrons.

